Projektbesprechung 12.1

Gruppe P1b

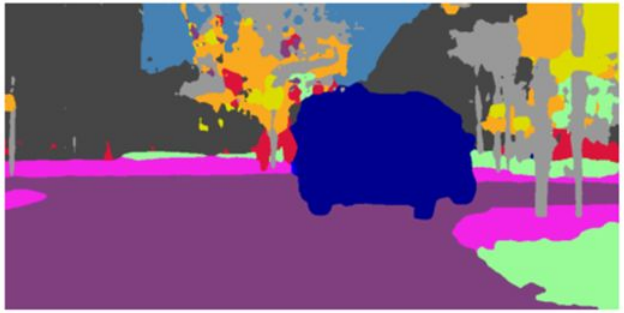
Training Classes

- **vegetation** und **parking** hinzugefügt

Original image

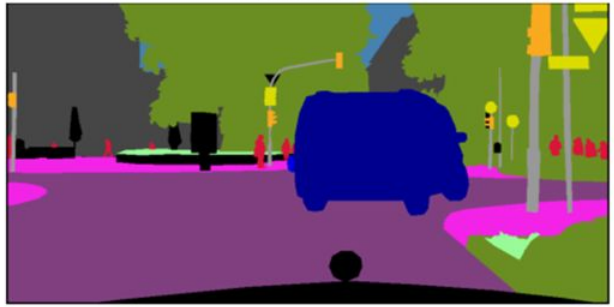


Predicted



vegetation auf 255 (ignored)

Ground truth



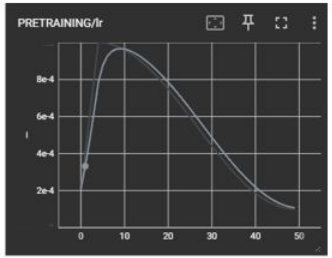
Predicted

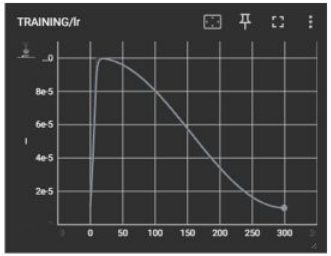


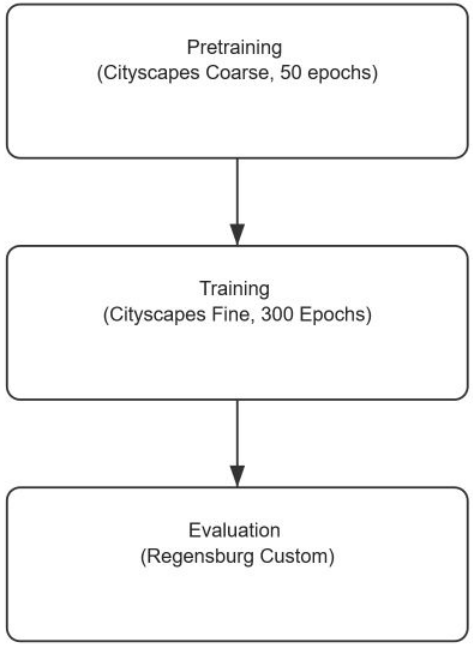
vegetation als lernbare Klasse

Training Setup

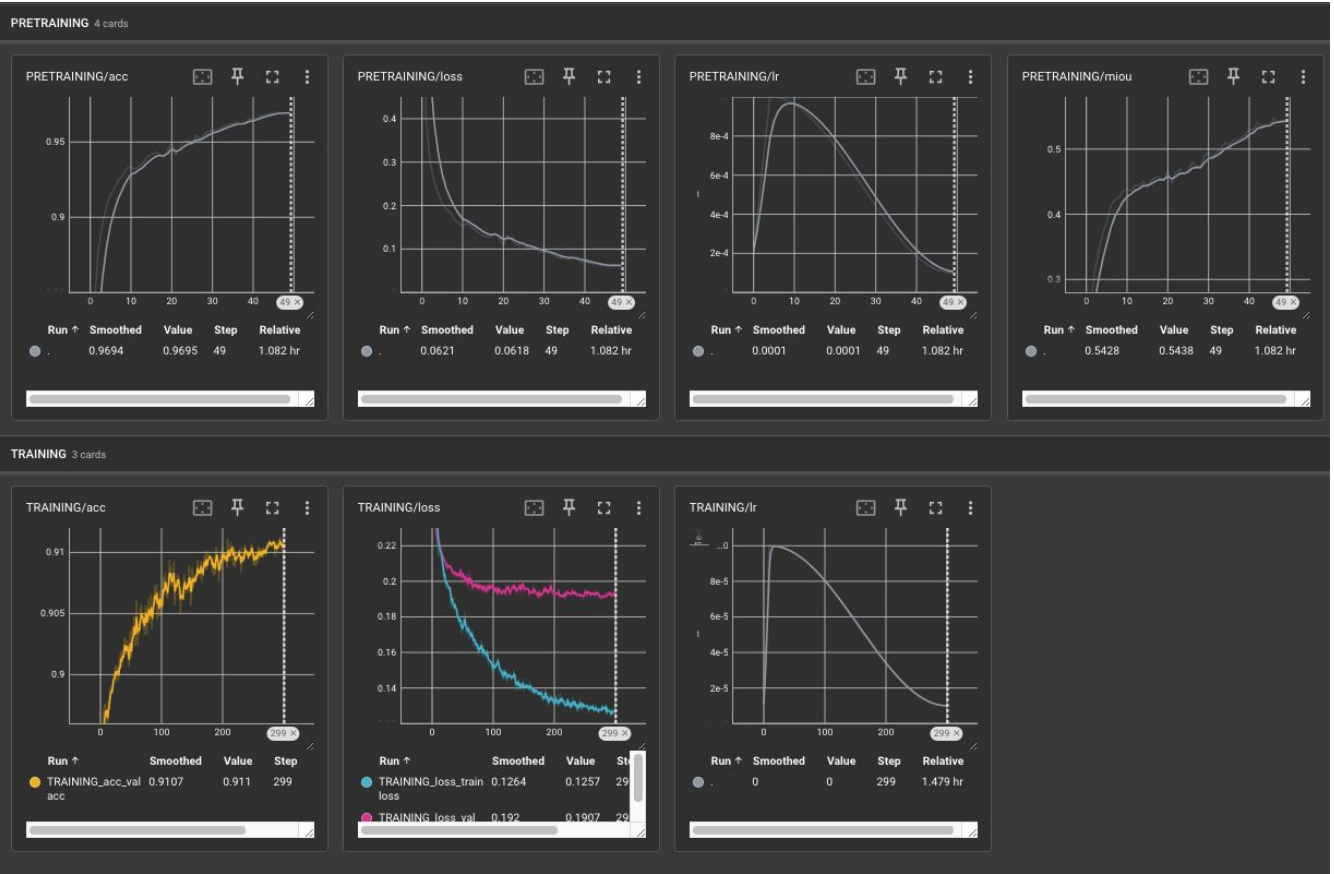
- **2-Stufen-Training**
  - Pretraining auf **coarse**
    - 50 epochs
    - 5 warmup epochs
    - learning rate (0.001 bis 0.0001)
  - Training auf **fine**
    - 300 epochs
    - 10 warmup epochs
    - learning rate (0.001 bis 0.00001)
  - Intersections degree = 3





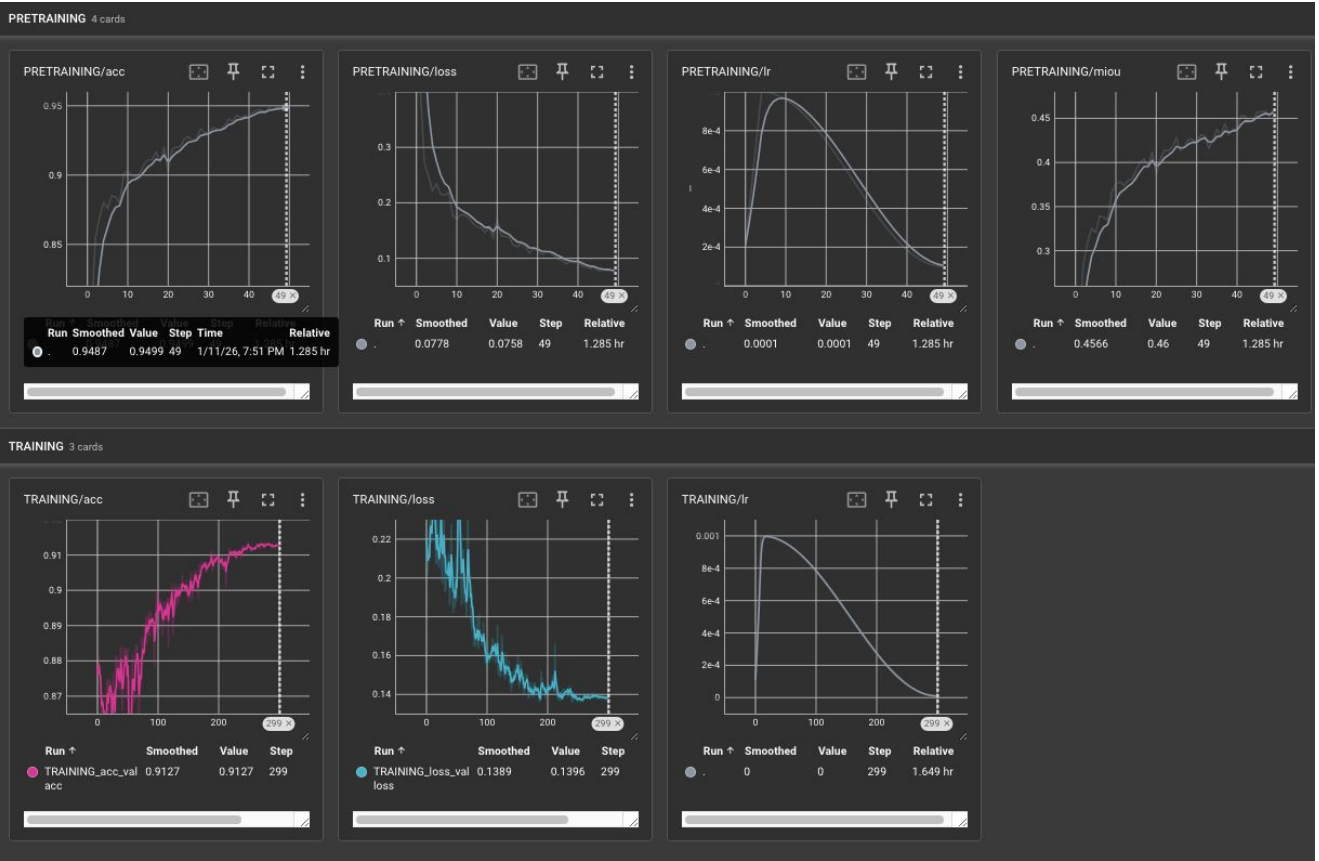


DeepLab



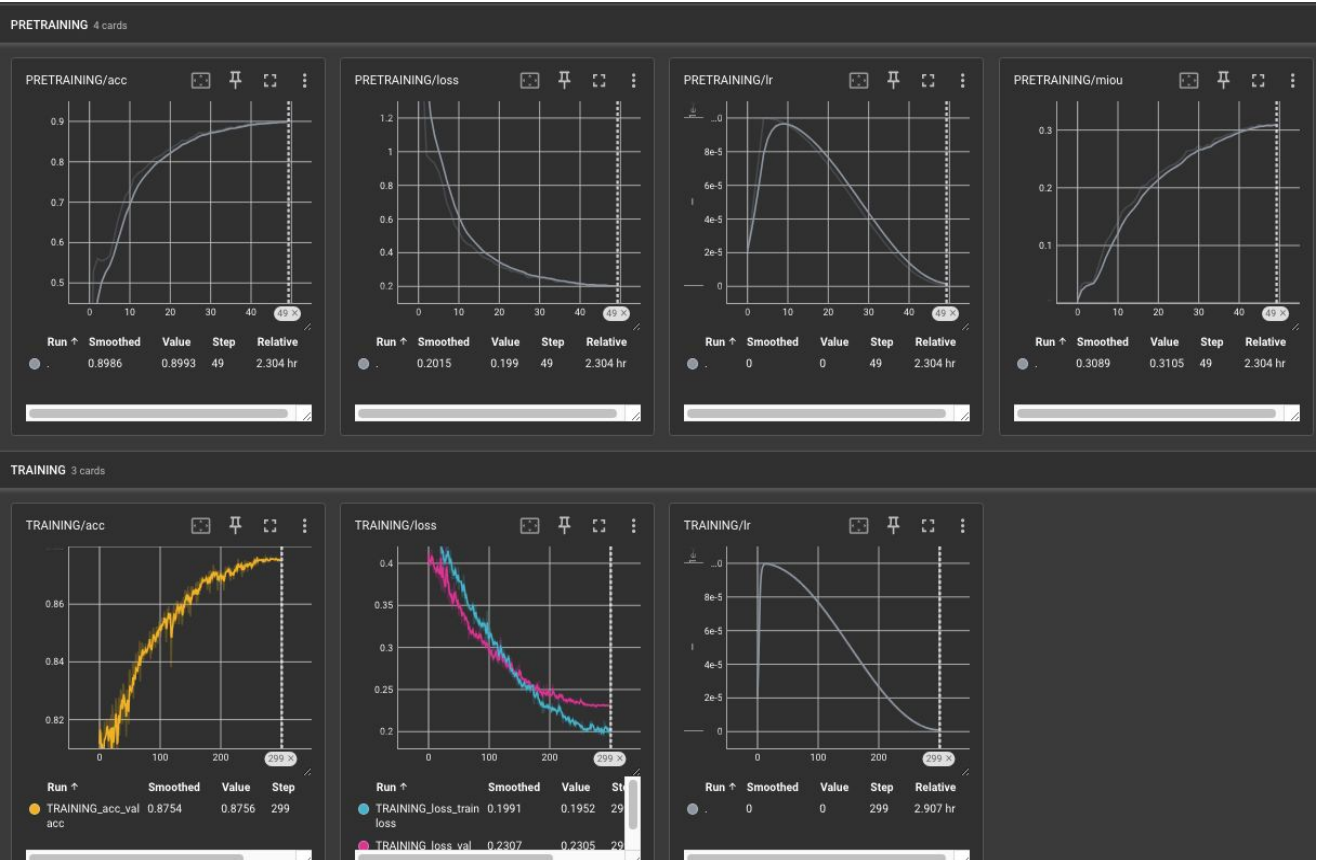
Train Loss=0.061, Val Loss=0.155, Train Acc=0.964, Val Acc=0.930,
Train MIoU=0.724, Val MIoU=0.501

Segformer



Train Loss=0.084, Val Loss=0.140,
Train Acc=0.935, Val Acc=0.913,
Train MIoU=0.580, Val MIoU=0.456

Unet



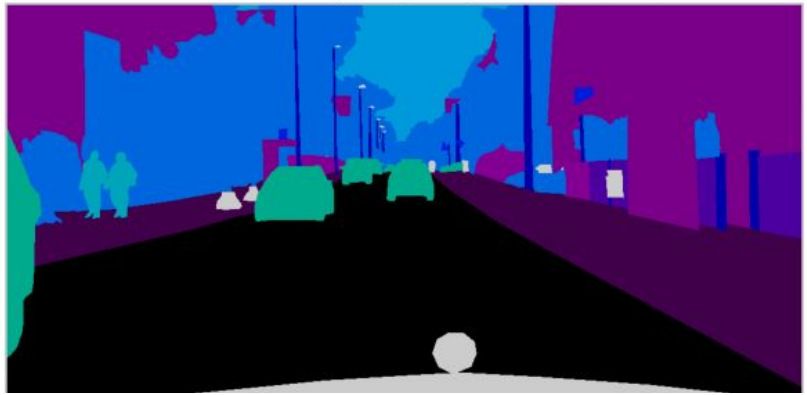
Train Loss=0.195, Val Loss=0.231,
Train Acc=0.887, Val Acc=0.876,
Train MIoU=0.366, Val MIoU=0.321

Augmentation Update

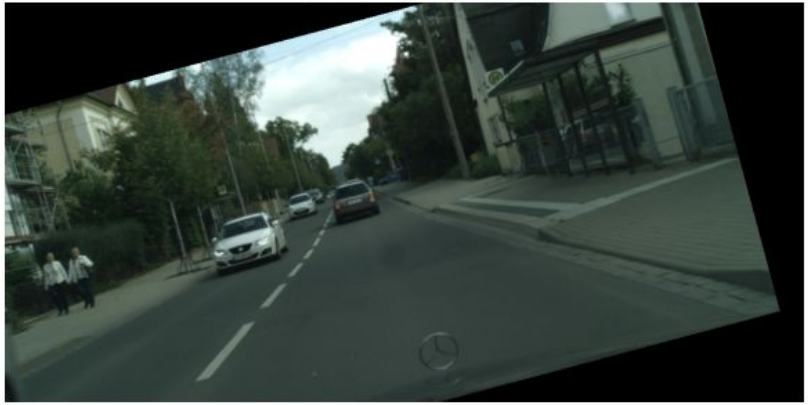
5. Color Jitter (Farbe) - Bild



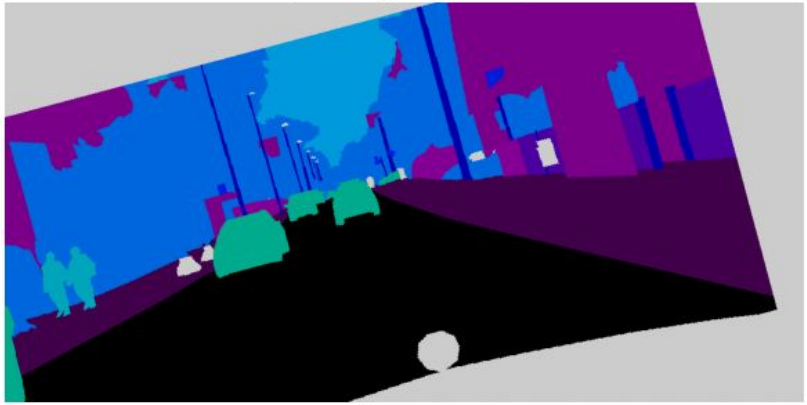
5. Color Jitter (Farbe) - Label



3. Affine (Drehung/Shift/Scale) - Bild



3. Affine (Drehung/Shift/Scale) - Label

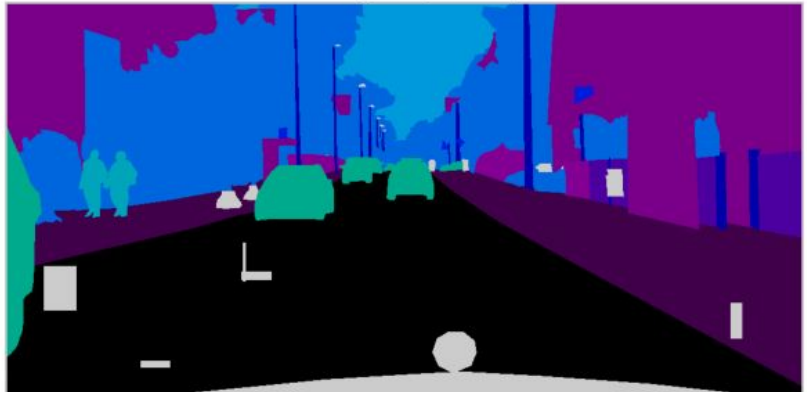


Augmentation Update

6. Coarse Dropout (Löcher) - Bild



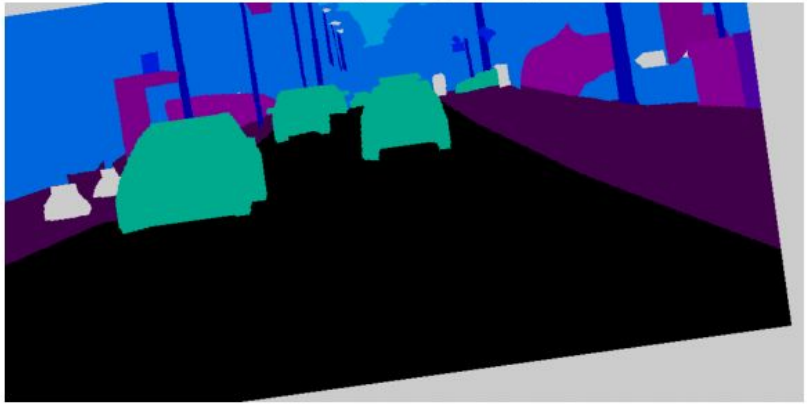
6. Coarse Dropout (Löcher) - Label



FULL PIPELINE (Training Config) - Bild

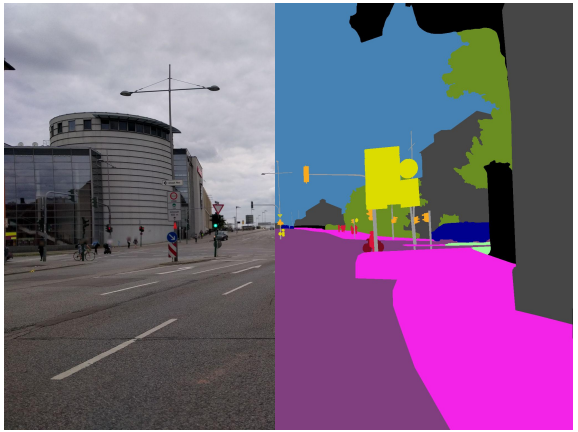


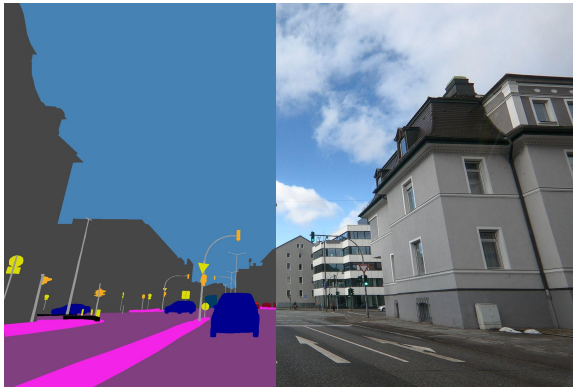
FULL PIPELINE (Training Config) - Label



Annotation abgeschlossen



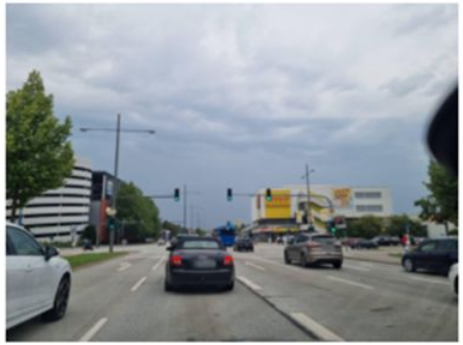






Evaluation (nicht final)

Original Image



Ground Truth



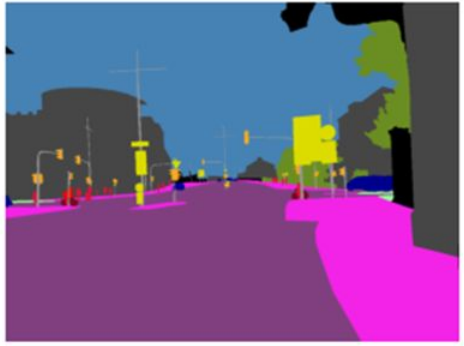
Predicted



Original Image



Ground Truth



Predicted



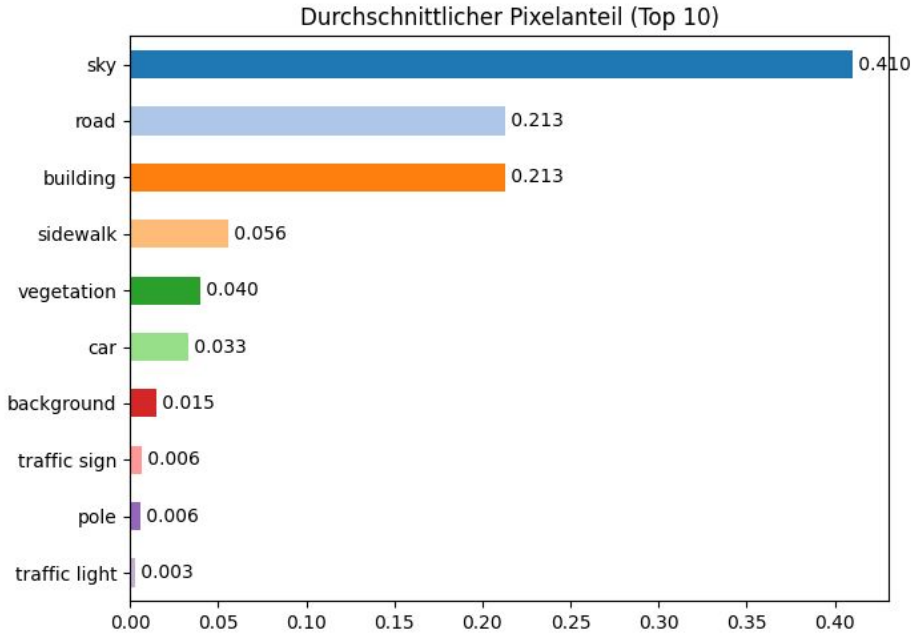
Extrahieren von Kennzahlen

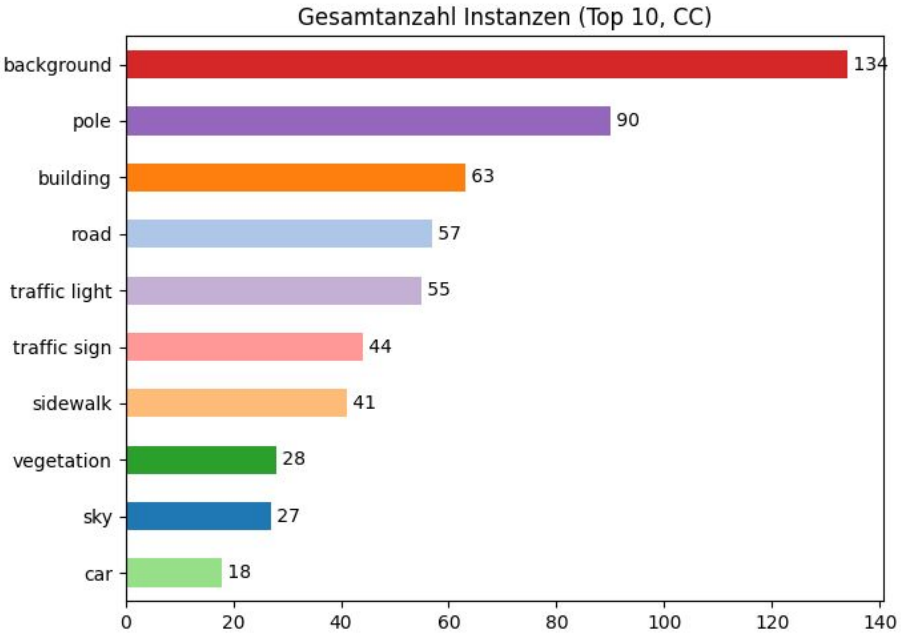
Mithilfe von Open CV mit 3 Methoden

- Instanzzaehlung: Connected Components (CC) als Baseline
- Alternative Instanzzaehlung: Watershed (Distance Transform + Seeds)
- Watershed Peak-Detektion

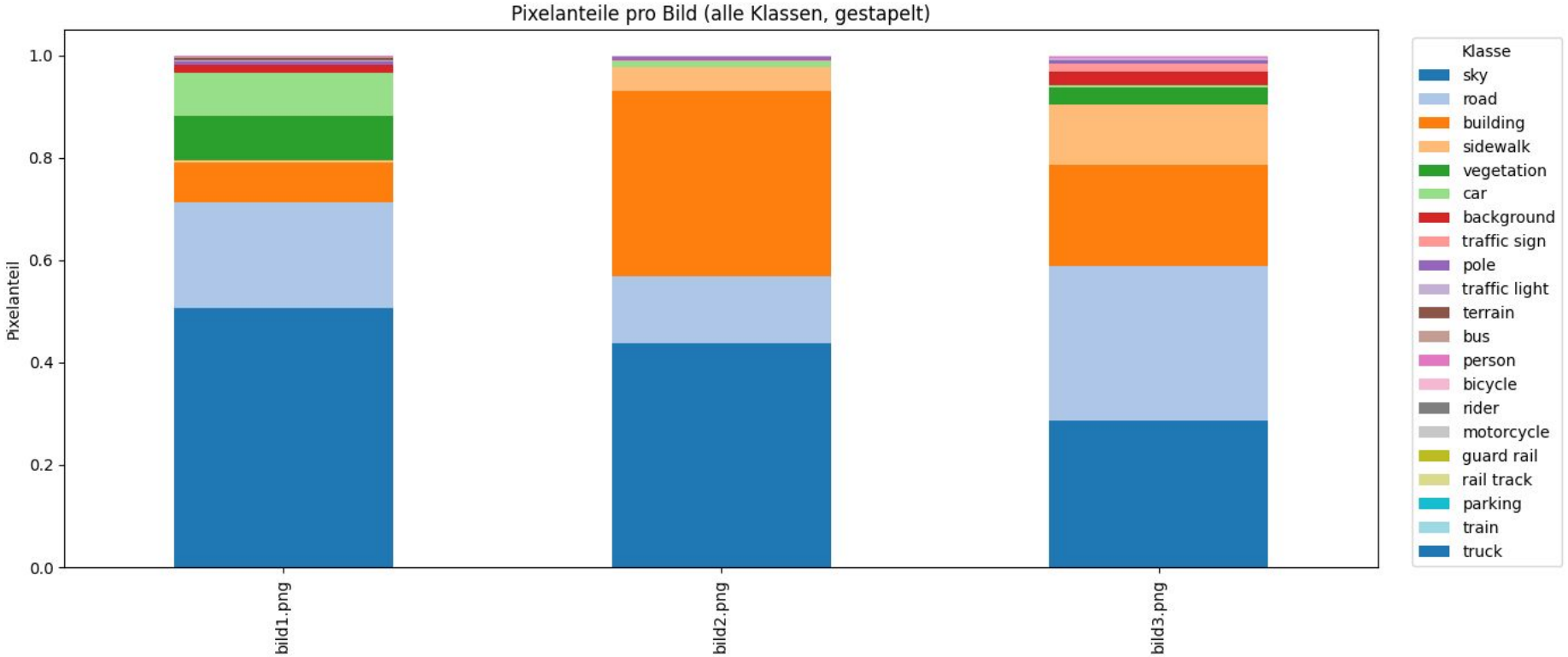
-> Test an 3 Bildern

Extrahieren von Kennzahlen

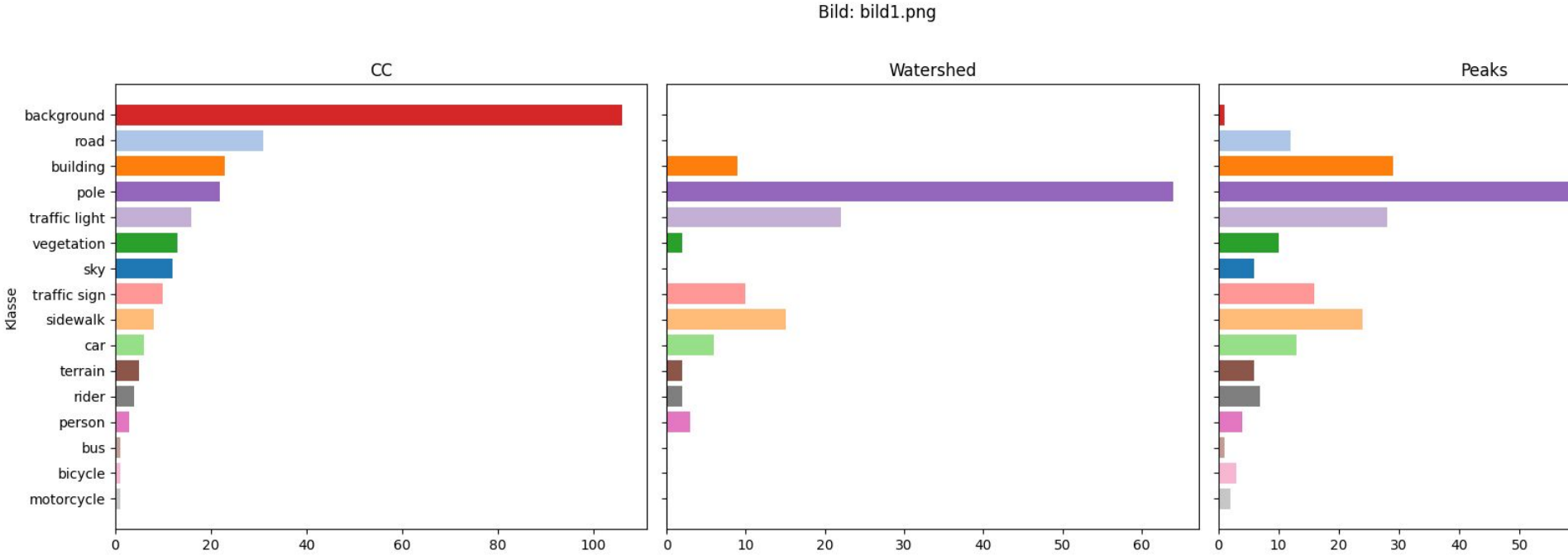




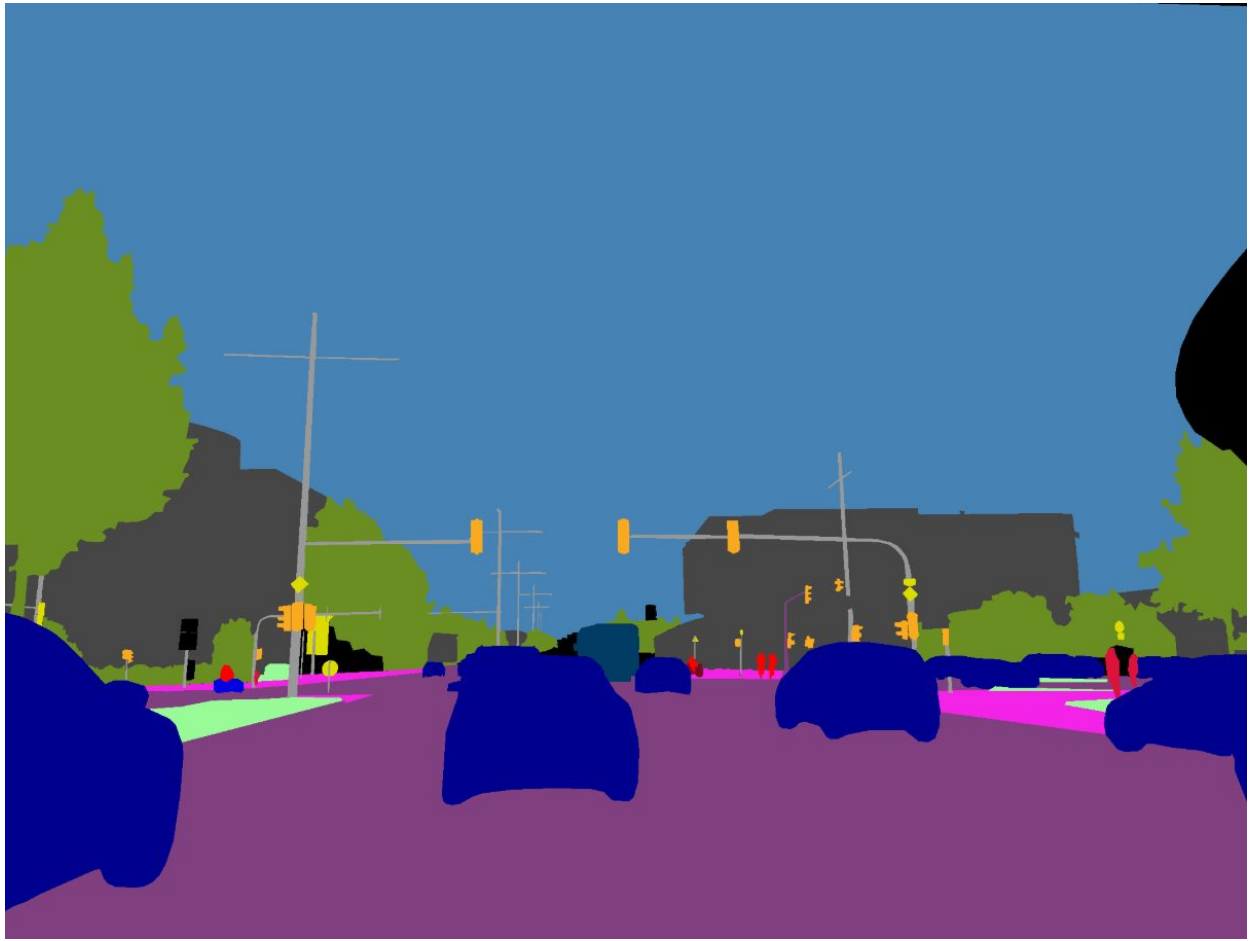
Extrahieren von Kennzahlen



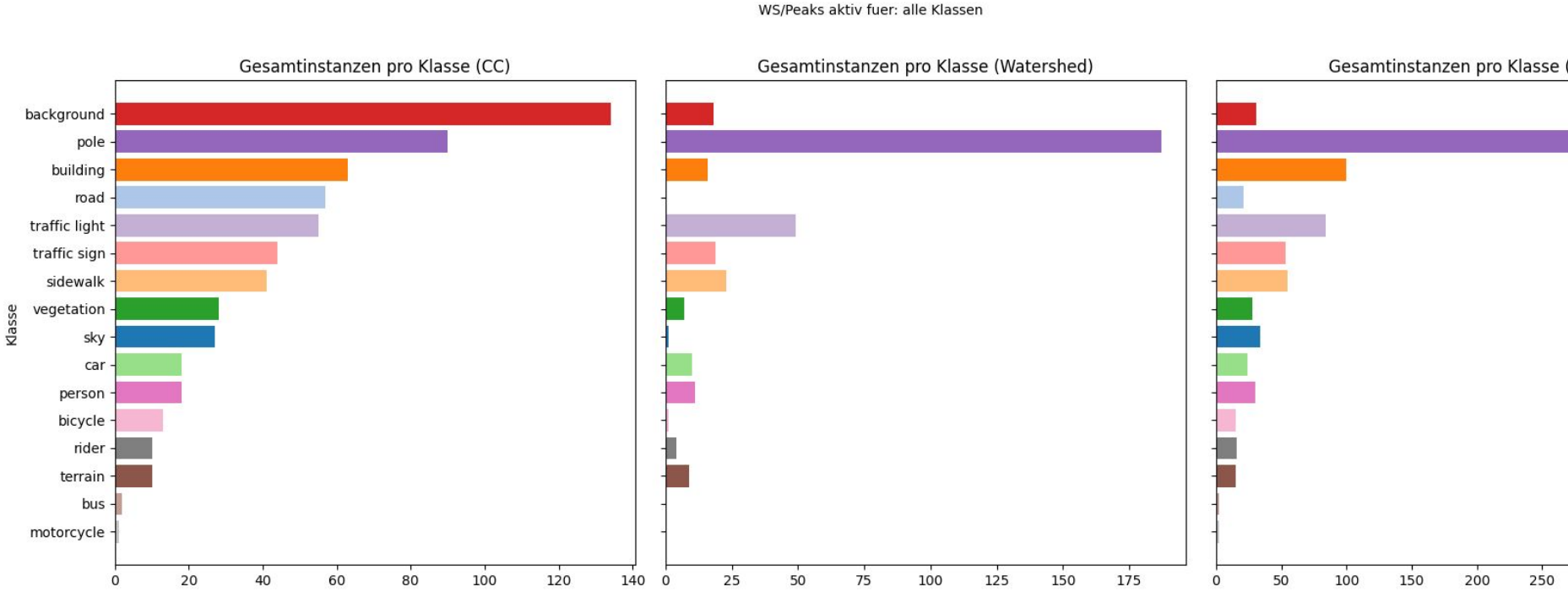
Extrahieren von Kennzahlen



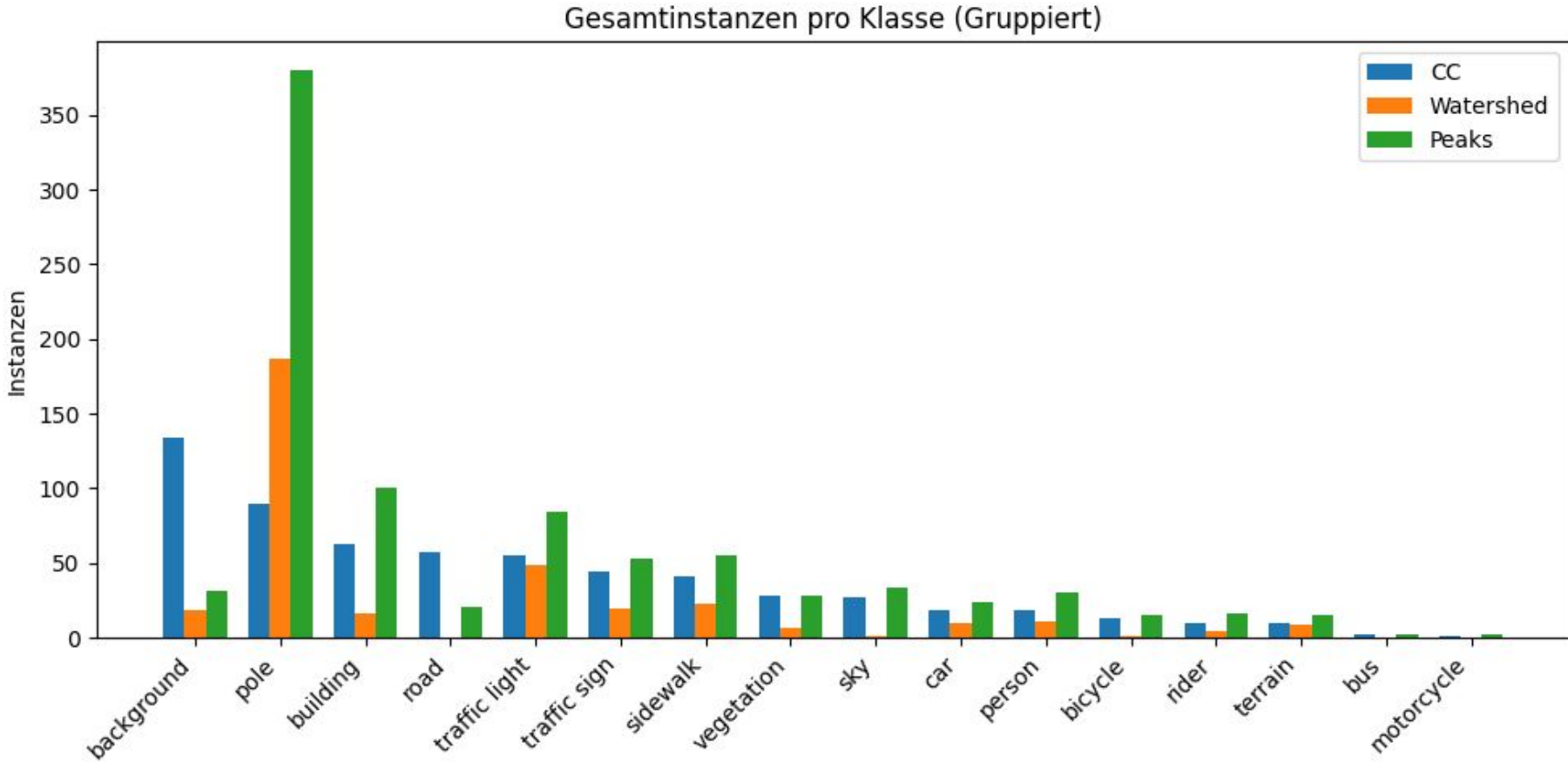
Extrahieren von Kennzahlen



Extrahieren von Kennzahlen



Extrahieren von Kennzahlen



Extrahieren von Kennzahlen

